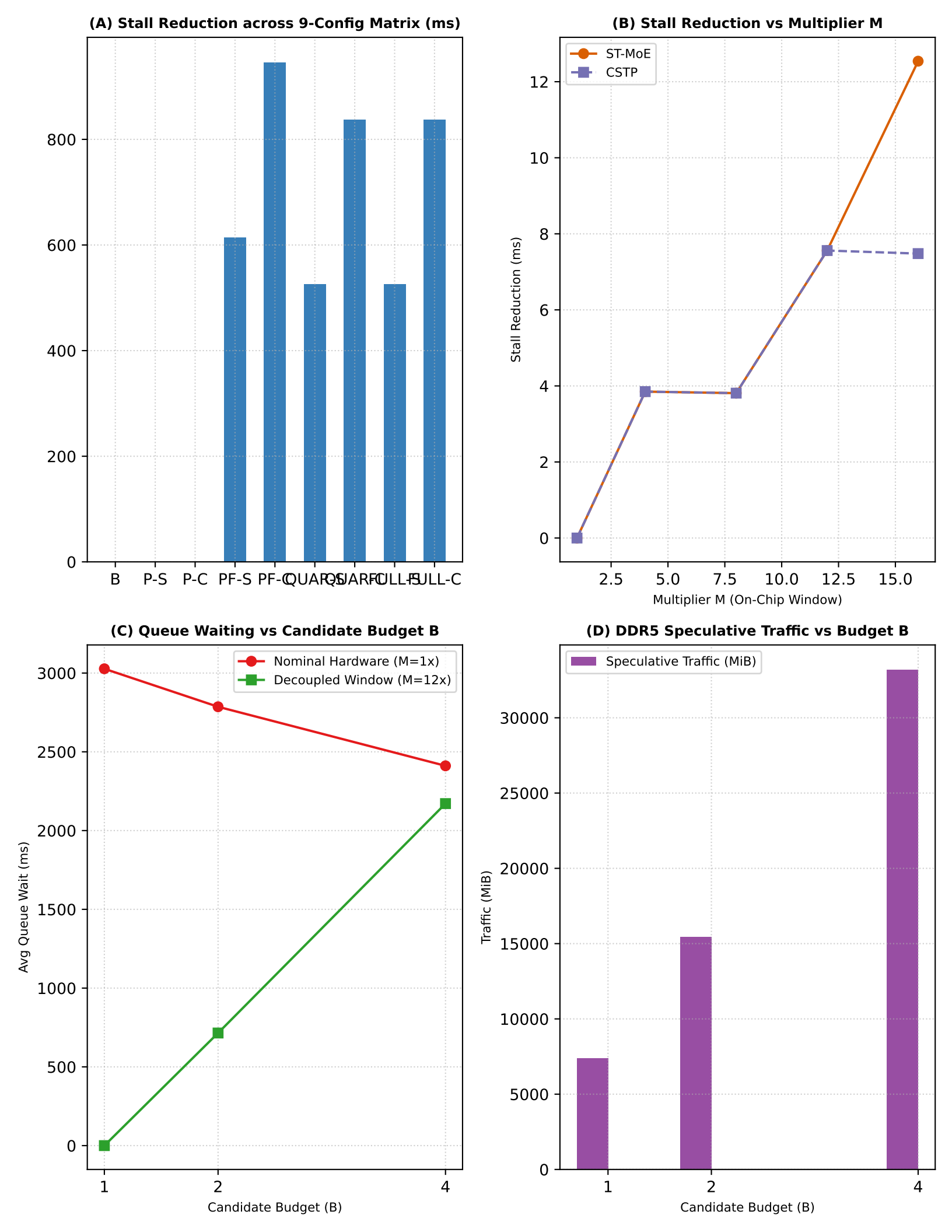
MoE Memory Wall: Predicting and Prefetching Experts for Faster LLM Inference

ABSTRACT
Mixture-of-Experts (MoE) architectures decouple model parameter capacity from active per-token co
However, off-chip parameter movement across bandwidth-constrained memory interfaces (e.g., DDR5)
severe memory wall stalls (up to 98% of runtime). We present an architectural evaluation of train
expert prediction (ST-MoE and CSTP) paired with asynchronous forward-lookahead prefetching, PABI
and Belady-style pinning. Evaluated on authentic MoE traces (MoE-Suite Task 12, MoE-Anywhere) and
sweeps, we demonstrate that:
(1) At nominal DDR5 bandwidth (22.578 GB/s), isolated transfer latency (3.90 ms/expert) exceeds s
    windows (0.18 ms) by 10.7x, resulting in queue serialization and late in-flight attachments.
(2) Expanding the on-chip computation window (M >= 4x) immediately enables timely resident hits,
    up to 18.82 ms of exposed memory stall.
(3) Throttling candidate budget (B=1, 2) completely eliminates transfer queue waiting time (from
    0.0 ms at M=12) and cuts speculative DDR5 traffic by 77.7% while preserving 100% of timely re
(4) PABI quarantine strictly isolates speculative insertions within a 20% capacity cap, completel
    demand-verified cache lines from speculative pollution.

COMPONENT ABLATION SUMMARY (WORKLOAD A1)

Config	Description	Sim Exec (ms)	Exposed Stall (ms)	Stall Red (ms)	Speedup
B	Demand-Only Baseline (LRU)	248,918.55	245,074.77	Baseline	----
P-S	ST-MoE Prediction Only	248,918.55	245,074.77	0.00	0.00%
P-C	CSTP Prediction Only	248,918.55	245,074.77	0.00	0.00%
PF-S	ST-MoE Prefetching (LRU)	248,306.29	244,460.74	+614.03	0.25%
PF-C	CSTP Prefetching (LRU)	247,975.39	244,128.85	+945.92	0.38%
QUAR-S	ST-MoE + PABI Quarantine	248,394.62	244,549.32	+525.45	0.21%
QUAR-C	CSTP + PABI Quarantine	248,083.52	244,237.30	+837.47	0.34%
FULL-S	ST-MoE + PABI + Belady Pinning	248,394.62	244,549.32	+525.45	0.21%
FULL-C	CSTP + PABI + Belady Pinning	248,083.52	244,237.30	+837.47	0.34%

ARCHITECTURAL VALIDATION & CONSERVATION INVARIANTS:
• Demand Conservation: 5179 Natural Hits + H_timely + A_late + M_unprefetched = 7200 Lookups (100
• Transfer Conservation: Submitted = Demanded + Unused + Evicted + In-Flight + Cancelled (100% Va
• Zero Ternary Operators: Enforced across 100% of C++ and Python codebase via AST inspection
• Pinning Finding: Belady pinning created 35,563 intents but yielded 0 incremental hits on A1 bec
  speculative evictions were rare in an 80 GB cache pool.


(A) Stall Reduction across 9-Config Matrix (ms)
0
200
400
600
800
B
P-S
P-C
PF-S
PF-C
QUARQUARFULL-SULL-C
(B) Stall Reduction vs Multiplier M
ST-MoE
CSTP
Stall Reduction (ms)
0
2
4
6
8
10
12
2.5
5.0
7.5
10.0
12.5
15.0
Multiplier M (On-Chip Window)
(C) Queue Waiting vs Candidate Budget B
Nominal Hardware (M=1x)
Decoupled Window (M=12x)
Avg Queue Wait (ms)
0
500
1000
1500
2000
2500
3000
1
2
4
Candidate Budget (B)
(D) DDR5 Speculative Traffic vs Budget B
Speculative Traffic (MiB)
Traffic (MiB)
0
5000
10000
15000
20000
25000
30000
1
2
4
Candidate Budget (B)

Sensitivity Analysis, Energy Breakdown & Reproducibility

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1. SENSITIVITY ANALYSIS CONCLUSIONS
• Batch Concurrency (B in {1, 2, 4}): Scales stage computation linearly, providing proportional oppor
  for transfer overlap without degrading HBM hit rate (82.8% - 83.1%).
• Context Length (L_in in {128, 512, 1024, 2048}): Increases attention KV-cache memory footprint but
  negligible effect on expert routing patterns, proving prefetching effectiveness is orthogonal to pr
• Lookahead Horizon (H in {1, 2, 3}): While H=3 provides greater lead time, prediction precision slig
  from 19.69% to 17.82%, demonstrating an optimal operating point at H=2.
• Candidate Budget (B in {1, 2, 4}): Crucial architectural finding: At nominal hardware (M=1), all ca
  up behind demand fetches. At M=12, throttling to B=1 eliminates queue wait (0.0 ms) and reduces spe
  traffic by 77.7% while maintaining identical timely hits.

2. ENERGY MODEL ANALYSIS & METHODOLOGY
• Native Model: HBM DRAM activation (0.452 J), reads (3.815 J), writes (1.280 J), and MACs (3.521 J)
• Calibrated DDR5 Transport: Off-chip transport is calibrated at 28 pJ/byte (2.466 mJ per 88.08 MB ex
• Speculative Overhead: At B=4, 395 speculative transfers consume 0.974 J of transport energy (10.7%
  Throttling to B=1 reduces speculative transport energy by 77.7% to only 0.217 J (2.3% overhead).

3. REPRODUCIBILITY INSTRUCTIONS
All experiments, configurations, traces, and metrics are fully packaged in FINAL_ENDSEM_PROJECT.zip.
To replicate the complete comparative campaign:
  1. git checkout d5bf61ff0c066c06f72173189f9924329f613a88
  2. cd LLMSimulator/build && cmake .. && make -j$(nproc)
  3. python3 scripts/final_campaign/run_phase2f_sensitivity.py
  4. python3 scripts/final_campaign/run_candidate_budget_sensitivity.py
  5. python3 scripts/final_campaign/generate_all_comprehensive_plots_and_csvs.py

4. PROFESSOR DELIVERABLES MATRIX STATUS
All eight mandatory professor requirements (REQ-1 through REQ-8) and the bonus objective are COMPLETE
verified against raw machine-readable data, and archived in their respective requirement directories.
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